Energy Through a Food Chain: Teacher Key

1. Meadow

20,000 kJ $\times 0.018 = 360$ kJ; $360 \times 0.12 = 43.2$ kJ; $43.2 \times 0.09 = 3.888$ kJ.

Overall fraction $= 3.888/20000 = 0.0001944$; overall percentage $= 0.01944\%$.

2. Pathways

A Pond: 375 kJ; 67.5 kJ; 9.45 kJ; 0.6615 kJ. Overall $= 0.00441\%$.

B Crop direct: 220 kJ; 55 kJ. Overall $= 0.55\%$.

C Woodland: 360 kJ; 54 kJ; 5.94 kJ; 0.4752 kJ. Overall $= 0.001584\%$.

3. Discovery

Adding gives 22.8%, which is nowhere near the overall efficiency. Decimal efficiencies: 0.018, 0.12, 0.09.

$$0.018 \times 0.12 \times 0.09 = 0.0001944.$$

Thus

$$\boxed{\eta_{\text{total}} = \eta_1 \eta_2 \eta_3 \cdots \eta_n}$$

when each η is expressed as a decimal fraction. Percentages cannot be added because each later efficiency applies to the already-reduced output of the previous stage, so the reference quantity changes at every transfer.

4. Direct calculation

Grass $\to$ vole $\to$ owl: $0.08 \times 0.13 = 0.0104 = 1.04\%$.

Phytoplankton $\to$ krill $\to$ penguin $\to$ seal: $0.16 \times 0.11 \times 0.09 = 0.001584 = 0.1584\%$.

Maize $\to$ insect $\to$ chicken $\to$ human: $0.07 \times 0.18 \times 0.12 = 0.001512 = 0.1512\%$.

5. Reverse cascade

$0.12\% = 0.0012$. If the third efficiency is x,

$$0.10 \times 0.15 \times x = 0.0012, \qquad x = 0.08 = 8\%.$$

The 90% loss at a nominal 10% transfer is 90% of the energy entering that particular trophic step, not 90% of the original energy supplied at the start of the chain.

6. Interpretation

The direct crop-to-human pathway has the greatest overall efficiency, 0.55%. A shorter chain avoids one or more multiplicative losses. Plausible causes of variation include differences in digestibility, metabolic rate, activity, body temperature regulation, fraction of prey actually consumed, fraction assimilated, quality of food, and environmental conditions.

Exit statement: "Successive efficiencies combine by *multiplication* because each percentage acts on *the output remaining from the previous stage.*"